

AMME5520 – Advanced Control & Optimization

Navigation & Control System

Project 2 - Report

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Design of a Navigation & Control System

Unmanned Hovercraft Vehicle

Introduction

This report encompasses the design process and conditions surrounding simulated control and navigation of an unmanned small hovercraft within a wind field.

$$M\ddot{q} + D(\dot{q} - \overline{V}_m(x, y)) = F$$

where,

$$q = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix}, F = \begin{bmatrix} (F_s + F_p) \cos \theta \\ (F_s + F_p) \sin \theta \\ l(F_s - F_p) \end{bmatrix}, M = \begin{bmatrix} m & 0 & 0 \\ 0 & m & 0 \\ 0 & 0 & J \end{bmatrix}, D = \begin{bmatrix} d_v & 0 & 0 \\ 0 & d_v & 0 \\ 0 & 0 & d_w \end{bmatrix}$$

with, q comprising the position (x, y) in metres and angle (θ) in radians, F the forces and torques, M the mass/inertia matrix, and D the damping matrix, and:

$$\overline{V}_w(x, y) = \begin{bmatrix} V_w(x, y) \\ 0 \end{bmatrix}$$

where, $V_w(x, y)$ is the wind vector at the point x, y as in Part 1.

The control inputs are port and starboard thrust F_p, F_s .

Design Specifications

In nominal forward motion, $F_p = F_s = 2$ and each has a maximum value of 2.5.

The parameters are mass $m = 5$, inertial $J = 0.05$, friction coefficients $d_v = 4$, $d_w = 0.4$ and $l = 0.1$.

Simulink Model

Non-Linear Dynamics

Question 1

The Simulink Model that simulates the nonlinear dynamics of the hovercraft and the effect of the wind field is highlighted below in *figure 1*.

Furthermore, the model assumes the beginning position or starting point of the hovercraft is from the origin point of the wind field, coordinates $(0,0)$ and is aimed to move towards the end points or limits of the wind field (i.e. move across the field), coordinates $(100, :)$.

The simulation currently is set to a simulation time of 125ms, however, can operate on any simulation time. The model has been configured with durable integrator and solver specifications.

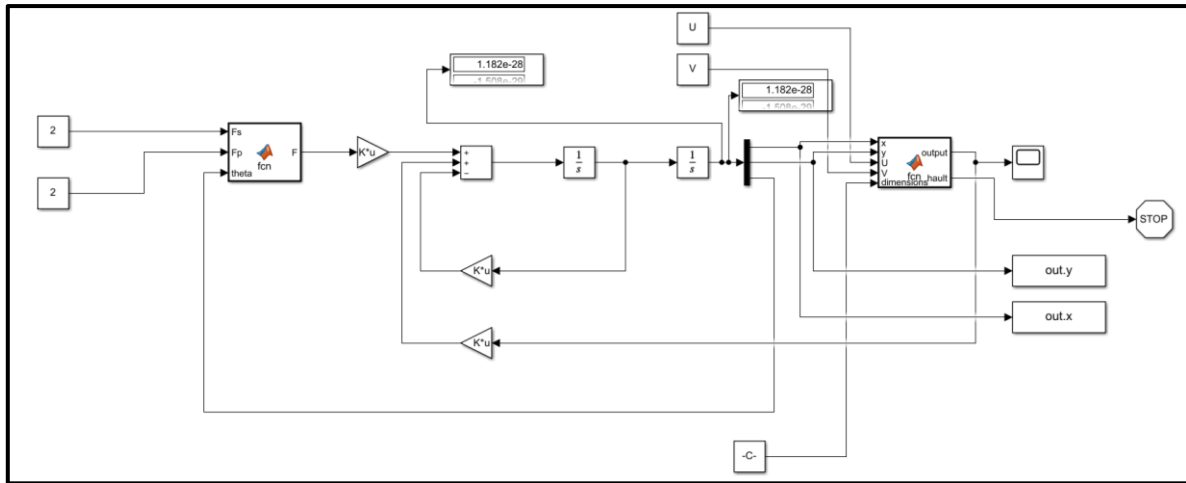


Figure 1 - Simulink Model (Q1)

The resulting path has been illustrated within *figure 2* and *figure 3*, where the specifications are set as the following:

- $F_s = 2$
- $F_p = 2$

Hence, both values are equal and within the range of the maximum value 2.5.

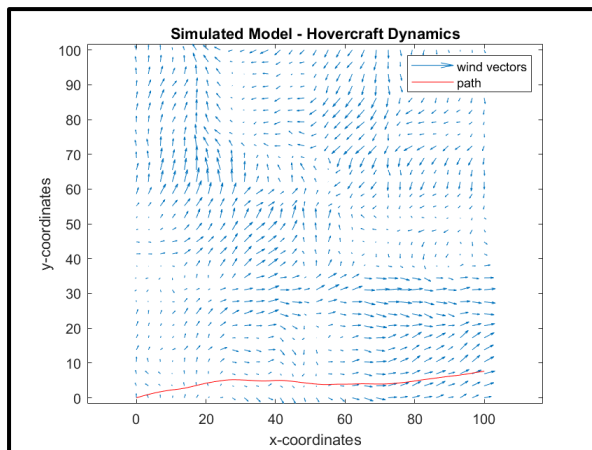


Figure 2 - Hovercraft Full Traversal of Field

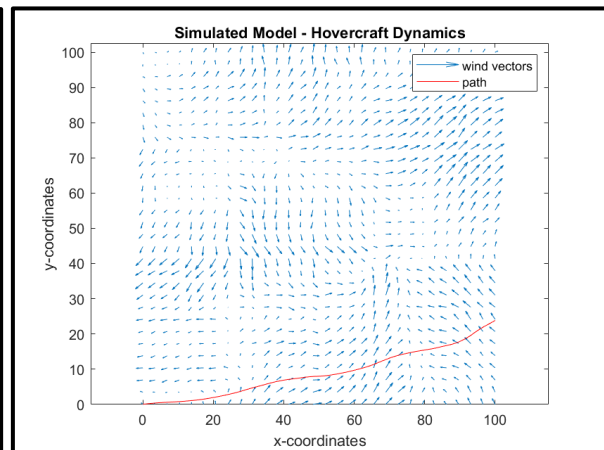


Figure 3 - Hovercraft Full Traversal of Field

There are various cases to which this model upholds its performance. If there is the case of the starting position already generating counter active wind effects or having wind directed in an opposing heading to the hovercraft path, then the simulated path of the hovercraft will not extend. This is accurate to reality upon which the hovercraft will be directed backwards. This is illustrated in *figure 4* to *figure 7*, displaying various possible generations of the map to which the model will denote such behaviour.

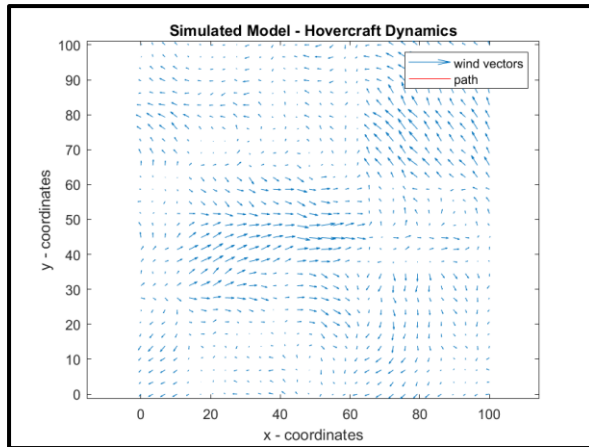


Figure 4 - Opposing Wind Start Situation

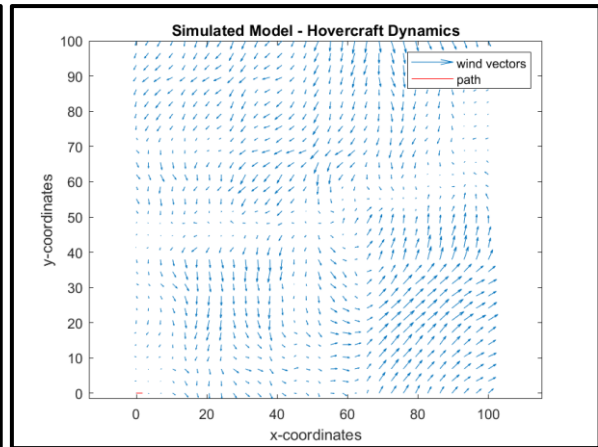


Figure 5 - Opposing Wind Situation (Gradual)

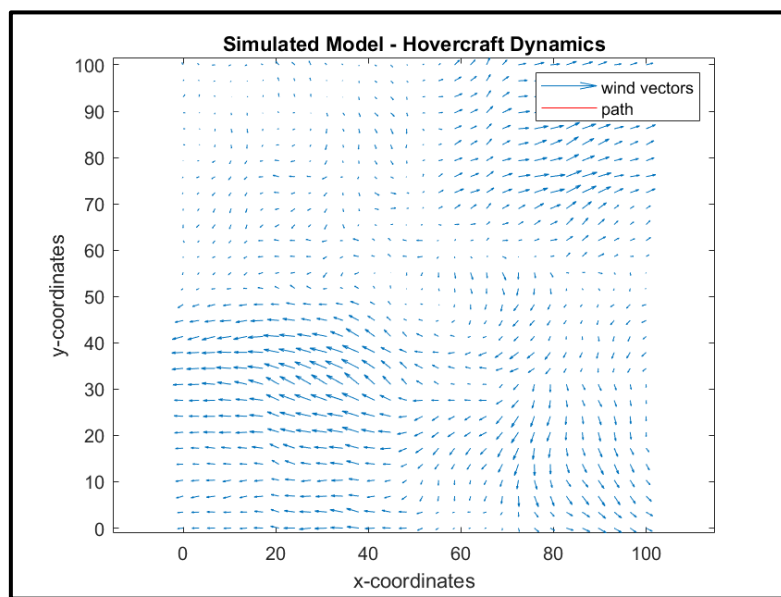


Figure 6 - Opposing Wind Start Situation

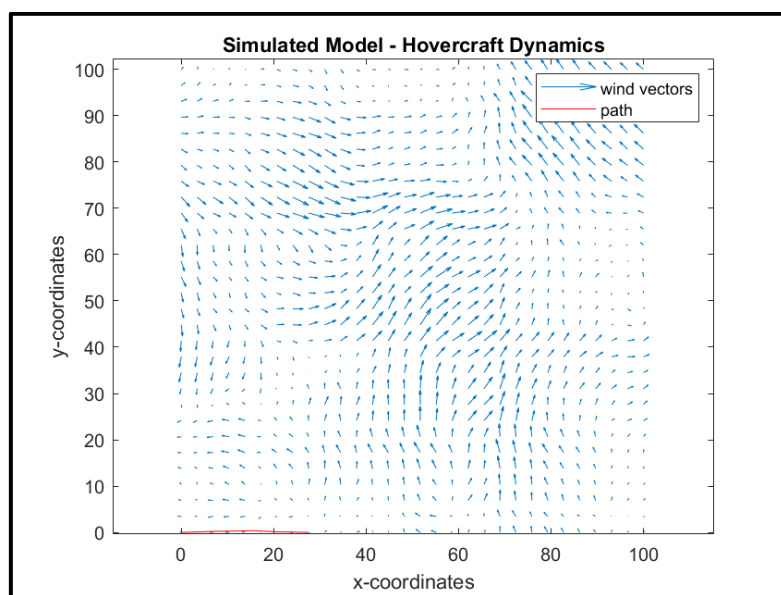


Figure 7 - Opposing Wind Situation (Gradual)

Furthermore, below is a display an alteration of the following specifications:

- $F_s = 2.5$
- $F_p = 2$

With a greater starboard thrust, in comparison to port thrust, the modelled dynamics of the hovercraft are illustrated below in *figure 8* and *figure 9*:

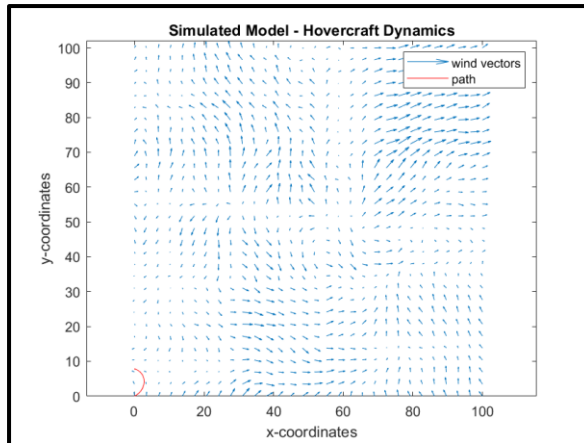


Figure 8 - $F_s > F_p$

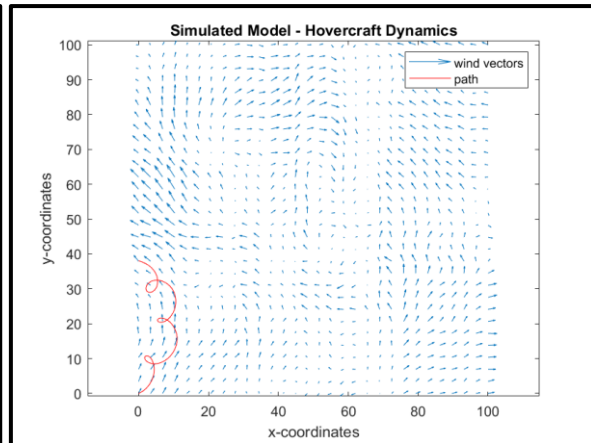


Figure 9 - $F_s > F_p$

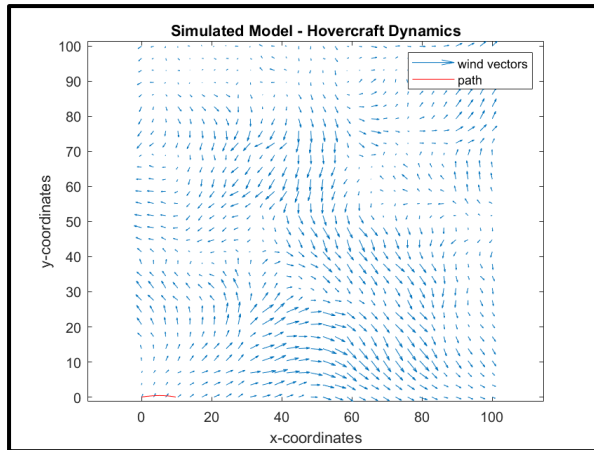
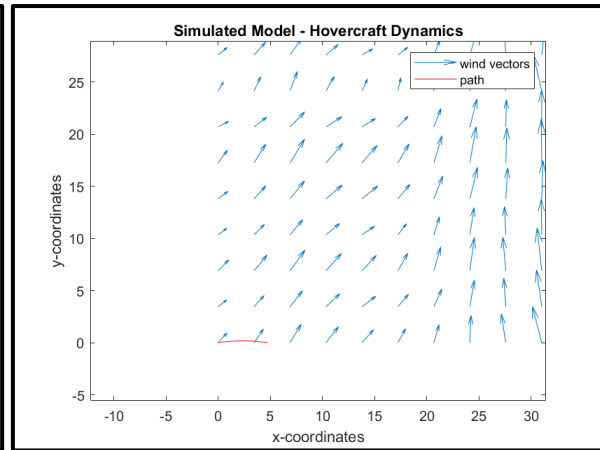
This movement is expected since a greater starboard thrust would result in an anti-clockwise rotation, and can be amplified in generations of the wind field complementing this movement.

Within the second example, it is highlighted that when the wind field either complements or opposes the dynamics of the hovercraft the path begins to perform full rotations and therefore, denotes accuracy with the model performance in relation to the wind field vectors.

As expected, the opposite is also true. This is displayed below within *figure 10* and *figure 11* when the following specifications are set:

- $F_s = 2$
- $F_p = 2.5$

Due to the wind effect existent near the starting point or origin upon various generations of the wind field, the dynamics of the hovercraft displays a near immediate downward movement. This has been highlighted through zooming in on the wind field and the expected clockwise rotation of the hovercraft can be seen.

Figure 10 - $F_p > F_s$ Figure 11 - $F_p > F_s$

A labelled version of the Simulink Model is denoted below:

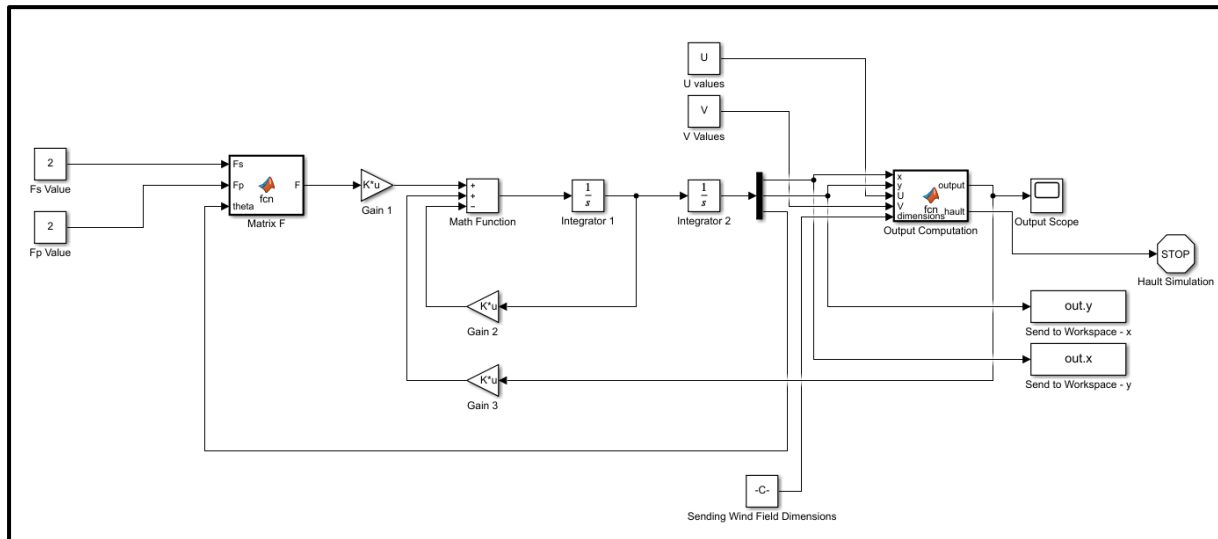


Figure 12 - Simulink Model (Final Version)

Linearisation of Dynamics

Question 2

$$M\ddot{q} + D\dot{q} - D\bar{V}_m(x, y) = F$$

For applications in state space model, we ignore $\bar{V}_m(x, y)$ therefore, resulting linearized equation approximation is in the form $\frac{dx}{dt} = f(x, u)$ where:

$$f(x, u) = -\frac{D}{M}\dot{q} + \frac{F}{M}$$

$$\frac{dx}{dt} = Ax + Bu$$

$$y = Cx + Du$$

Note:

$$x_1 = \dot{q} \text{ \& } x_2 = \ddot{q}$$

So, the following section below highlights the computations of x_1 & x_2 respectively;

The equation is linearized by taking the partial derivative of the right-hand side of the equation for both x and u :

$$A_{x_1} = \frac{\partial \left(-\frac{D}{M} \dot{q} + \frac{F}{M} \right)}{\partial x} = -\frac{D}{M} + \frac{F}{M}$$

For expression $\frac{F}{M}$ we must introduce value E as a consideration of the variable θ within $\frac{\partial f}{\partial u}$.

$$\therefore F = Eu \rightarrow E = \frac{F}{u}$$

$$B_{x_2} = \frac{\partial \left(-\frac{D}{M} \dot{q} + \frac{Eu}{M} \right)}{\partial x} = \frac{E}{M}$$

$$\frac{dx}{dt} = 0 + -\frac{D}{M}(x - x_{ss}) + \frac{E}{M}(u - u_{ss})$$

For $x_2 = \ddot{q}$ we can conclude that assuming $x = g(x, u) \rightarrow \dot{x} = 0$ due to the eliminate of u control input.

$$A_{x_2} = \frac{\partial \left(-\frac{D}{M} \dot{q} + \frac{F}{M} \right)}{\partial x} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$B_{x_2} = \frac{\partial \left(-\frac{D}{M} \dot{q} + \frac{Eu}{M} \right)}{\partial x} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

The linearized differential equation that approximates $\frac{dx}{dt} = f(x, u)$ is the following:

$$\frac{dx}{dt} = f(x_{ss}, u_{ss}) + \left. \frac{\partial f}{\partial x} \right|_{x_{ss}, u_{ss}} (x - x_{ss}) + \left. \frac{\partial f}{\partial u} \right|_{x_{ss}, u_{ss}} (u - u_{ss})$$

Therefore, the final linearisation of the dynamics along a trajectory is expressed as the following:

$$f(x', u') = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \\ \ddot{x} \\ \ddot{y} \\ \ddot{\theta} \end{bmatrix} = Ax' + Bu'$$

$$= \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & -(F_s + F_p) \sin \theta & -\frac{d_v}{m} & 0 & 0 \\ 0 & 0 & (F_s + F_p) \cos \theta & 0 & -\frac{d_v}{m} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{d_w}{J} \end{bmatrix} \begin{bmatrix} x' \\ y' \\ \theta' \\ \dot{x}' \\ \dot{y}' \\ \dot{\theta}' \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ \frac{\cos \theta^*}{m} & \frac{\cos \theta^*}{m} \\ \frac{\sin \theta^*}{m} & \frac{\sin \theta^*}{m} \\ \frac{L}{J} & -\frac{L}{J} \end{bmatrix} \begin{bmatrix} F_s' \\ F_p' \end{bmatrix}$$

Controllability

From the linearisation of the dynamics, it is evident that the system is controllable, evident by the control input u .

Control System

Question 3

Integration with *Project Part 1* as well as improvement upon the Dijkstra's Algorithm implementation allowed for the results shown in *figure 13*, hence, confirming operation of the shortest path being able to be determined or calculated in the model.

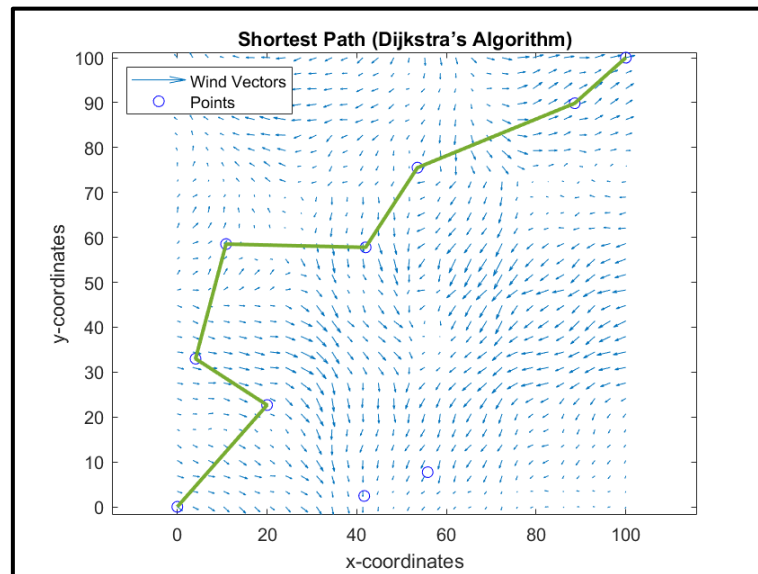


Figure 13 - Dijkstra's Algorithm Exemplar Operation

Therefore, it can be seen that the hovercraft begins at origin point and desires to reach coordinates (100,100). With an increase of sample points representing the various locations that are being evaluated for their cost, *figure 14* and *figure 15* captures this case and displays its performance.

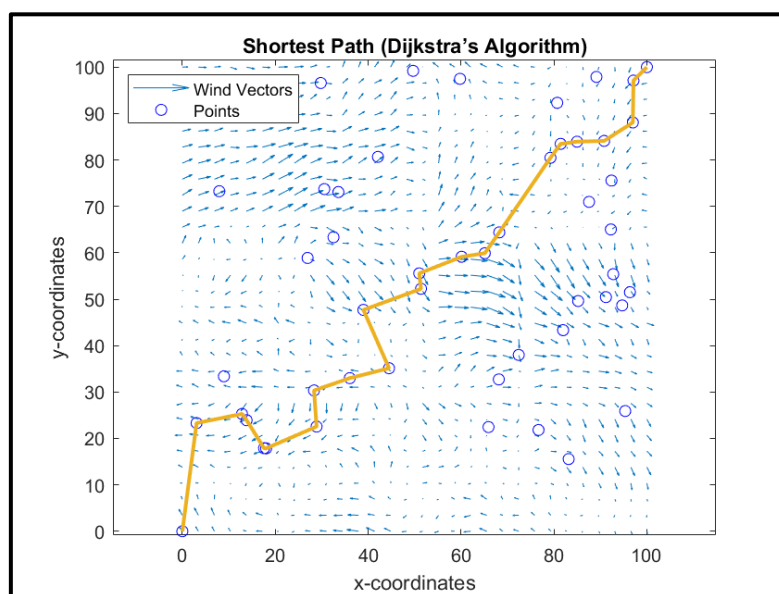


Figure 14 - 50 Point Exemplar Operation

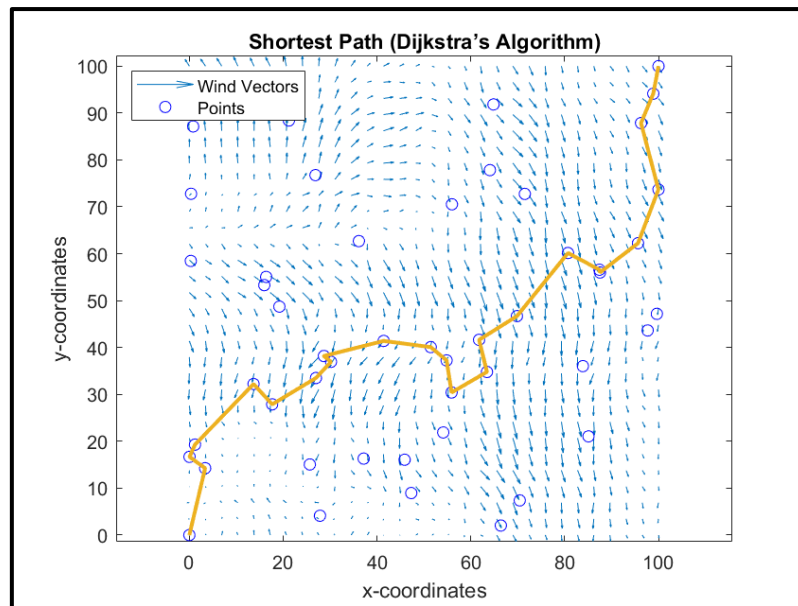


Figure 15 - 50 Point Exemplar Operation

Furthermore, below *figure 16* denotes the Simulink Model for the control system of the hovercraft in order to control the hovercraft along the paths found within *figure 15* above.

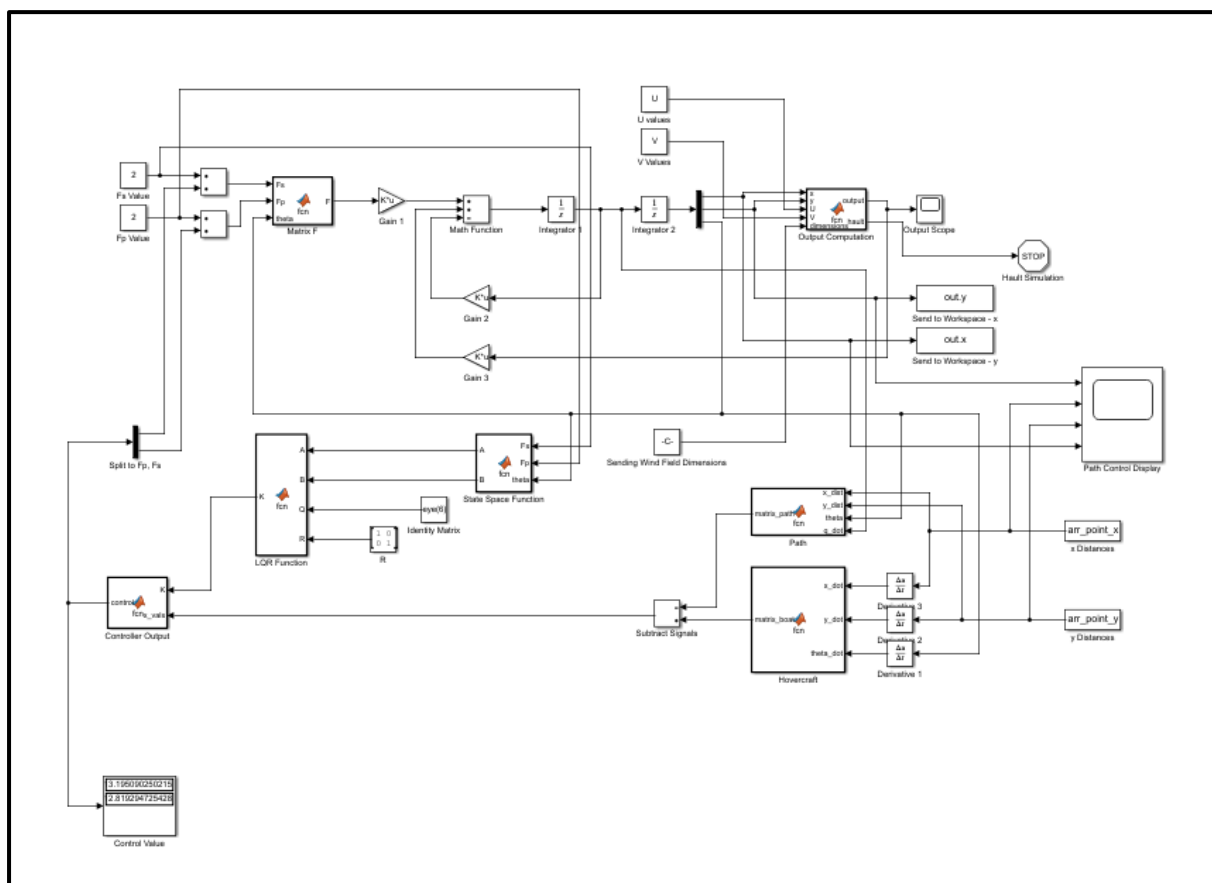


Figure 16 - Simulink Model Control System

The design process involves implementing the state space model found within the previous section. More specifically two functions process the distance over the total path being sent into the path function and hence, the derivative of this represents the movement of the hovercraft. Similarly, we

must utilize time either from the cost function in *Project Part1*. An LQR is then set up to process the state space model $Ax + Bu$ and a participant configurable R value. Hence, allowing the control of the hovercraft over the paths.

The calculation of the K value is critical to creation of this control system model, which was completed primarily utilizing the LQR function.

The step function incorporates taking various samples of the original path created and hence, allowing for an accurate control over the x and y movement of the hovercraft. The x and y control in one such example is displayed below in *figure 17*.

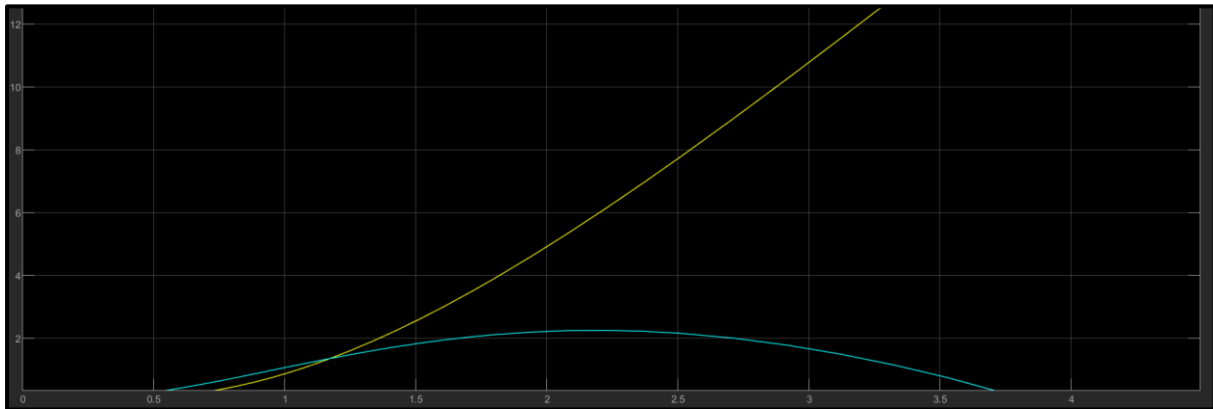


Figure 17 - Step Function

Controller Design

Question 4

The design of the controller that incorporates the case of the wind vector being constant in an unknown direction acting along the straight-line segment can still be employed with majority of the design aspects of the control system in the above section.

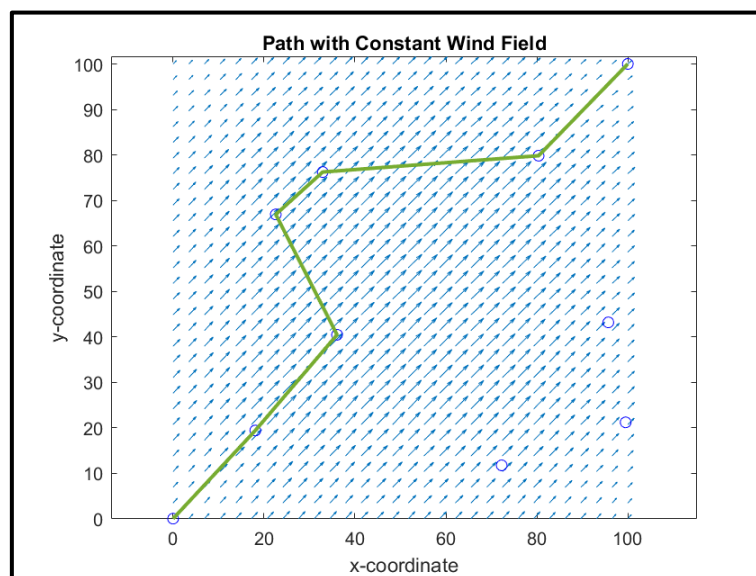


Figure 18 - Path when Constant Wind Field

The operation of the control system would still require the implementation of an integration controller to minimize the steady state error. Therefore, the implementation of a Proportion-Integral-Derivative (PID) controller would allow for the minimization of the steady-state error.

The PID controller is selected as it is understandable to participants of the system and effective in feedback compensation. Furthermore, it is functionable and understood conceptually with differentiation operations such as those existent in this system. The simplified design that is to be implemented onto the Simulink Controller Model is denoted in *figure 19*.

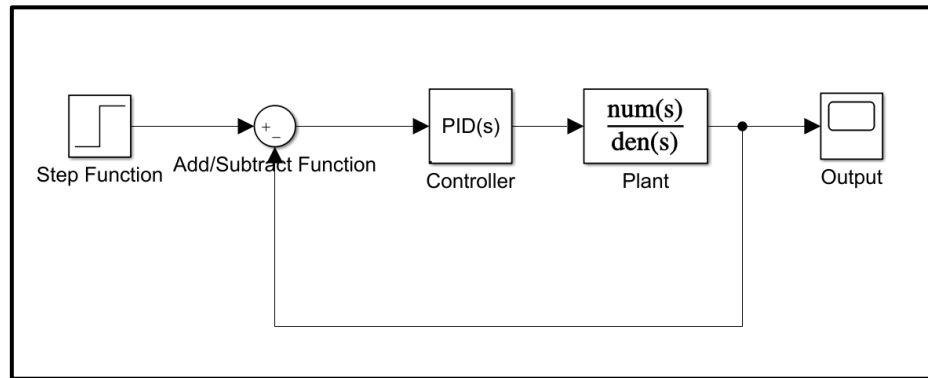


Figure 19 - PID Pseudo Design

The following *figure 20* highlights the PID controller design effect, into the Simulink Model from the previous section, upon the output. The step function is the defined r value within the PID controller which consequently experiences a reduction of the steady-state error.

With implementation of upon the previous Simulink Model control system, it can be evident that through manipulation of the K_p , K_i and K_d and tuning with solver configuration, the response is illustrated below.

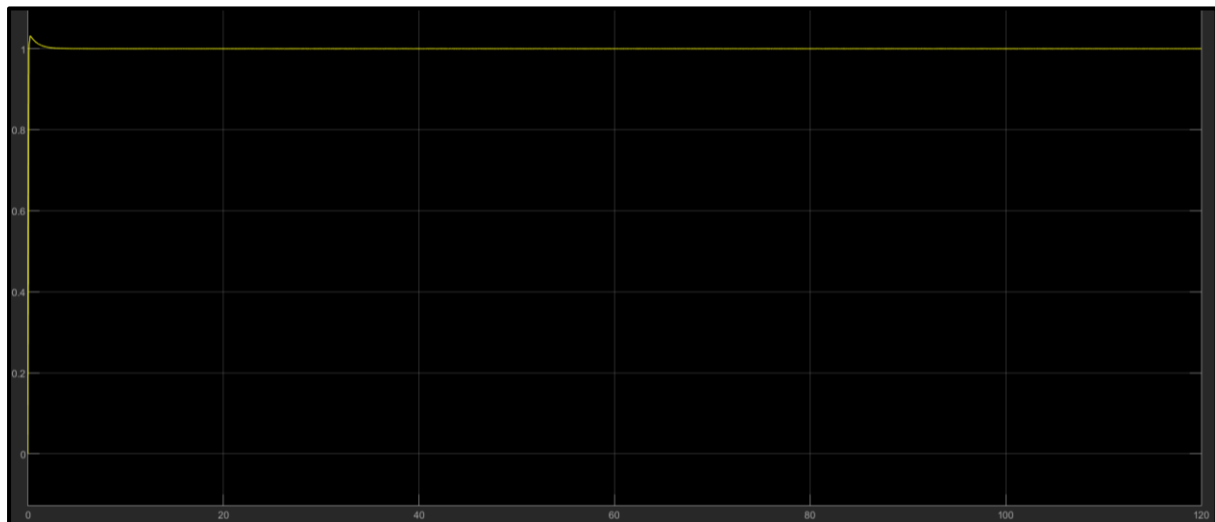


Figure 20 - PID Control Design Performance (Step Response)

Kalman Filter

Question 5

EKF Design

Question 6